Collaborative Overtaking Strategy for Enhancing Overall Effectiveness of Mixed Connected and Connectionless Vehicles

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***Abstract*—Intelligent Transportation Systems (ITS) aim to enhance traffic management by improving connectivity and data sharing among vehicles and road infrastructure. In a Mixed Connected and Connectionless Vehicles (MCCV) scenario consisting of connected vehicles equipped with On-Board Units (OBUs) and non-connected vehicles lacking OBUs, communication disparities create challenges in critical lane-changing overtaking decisions. These discrepancies hinder the adaptation of fully connected scenarios to dynamic interactions among these different types of vehicles. Considering the diversity in decision-making ways and capabilities of non-connected vehicles in MCCV scenarios, ensuring the coordinated execution of safe and efficient lane-changing overtaking maneuvers by multiple connected vehicles is crucial for enhancing traffic efficiency. Therefore, we propose a collaborative strategy to facilitate safer and more efficient lane-changing overtaking maneuvers for connected vehicles in the MCCV scenario. First, we design a multi-criteria priority detection, and a dynamic event-triggered mechanism based on confidence intervals to foster efficient collaboration among connected vehicles, optimizing decision-making and reducing conflicts. Second, to accommodate diverse driving styles of autonomous and human-driven vehicles, we introduce an Improved Dynamic Precise Fuzzy C-Means (IDP-FCM) algorithm to dynamically identify and adapt to different driving styles, thereby improving safety. Finally, tackling the challenge of multiple connected vehicles performing lane-changing overtaking involving hybrid action space, our proposed Multi-agent Contrastive Parameterized Dueling Deep Q-Network (MCPDDQN) algorithm incorporates contrastive learning to improve strategy stability in complex driving scenarios. Experimental results demonstrate the effectiveness of our strategy in improving road safety and traffic efficiency of the MCCV scenario.**

***Index Terms*—Collaboration, driving style, ITS, lane-changing overtaking strategy, MCCV, OBUs.**

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I. Introduction

With the increasing number of vehicles and demand for road traffic, the rapid advancement of Intelligent Transportation Systems (ITS) and autonomous driving technology presents innovative solutions to the challenges of traffic efficiency and road safety [1]. Vehicles equipped with On Board Units (OBUs) can connect and share environmental information such as location and speed with surrounding connected vehicles and road infrastructure [2]. This advanced level of automation and collaborative driving is expected to enhance road safety and overall traffic efficiency. Yet, high costs associated with technology and infrastructure development will hinder the rapid adoption of connected vehicles, ushering in a transitional phase of Mixed Connected and Connectionless Vehicles (MCCV). This MCCV environment involves complex interactions between connected vehicles and surrounding non-connected vehicles, especially in high-density and dynamically changing traffic situations. Lane-changing overtaking maneuvers, as an important means for vehicles to increase their speed, play a pivotal role in augmenting road traffic efficiency and safety. However, improper execution of these maneuvers not only escalates the risk of traffic accidents but also impairs the overall efficiency of traffic flow. Data from the National Highway Traffic Safety Administration in the United States indicate that lane-changing overtaking maneuvers are responsible for 27% of all traffic accidents [3]. Consequently, ensuring that connected vehicles can effectively collaborate and make safe and efficient lane-changing overtaking decisions in the MCCV scenario becomes a significant research area.

Existing studies have achieved certain progress in lane-changing overtaking by Connected Autonomous Vehicles (CAVs) [4]. However, the majority of these studies concentrate

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on individual vehicle benefits, such as congestion avoidance or speed enhancement, at the expense of considering the dynamics of vehicle interactions and the potential for competitive conflicts arising from a lack of collaboration [5]. To overcome the limitations of the single-agent paradigm, researchers propose multi-vehicle collaborative lane-changing overtaking decisions based on a multi-agent paradigm, improving overall traffic efficiency and safety through information sharing and cooperation among vehicles [6]. In the persistent MCCV scenario, with an increasing ratio of connected to non-connected vehicles owing to technological progress, focusing solely on the autonomous decisions of connected vehicles may precipitate decision-making conflicts among them driven by self-interest. For instance, in scenarios with three lanes, simultaneous attempts by vehicles on both the left and right lanes to merge into the middle lane for speed gains could lead to resource contention and decision-making conflicts, elevating the risk of safety incidents. Integrating collaborative strategies among connected vehicles with individual decision-making processes could mitigate such conflicts. However, the absence of communication capabilities in non-connected vehicles makes the direct application of fully connected autonomous driving strategies based on a multi-agent approach in the MCCV scenario unfeasible. Consequently, achieving collaborative lane-changing overtaking among connected vehicles within MCCV environments represents a substantial challenge.

The diversity of driving styles among vehicles on the road, encompassing both human-driven and Autonomous Vehicles (AVs), plays a crucial role in influencing road safety and optimizing traffic flow. This diversity stems from a variety of factors, including differences in individual drivers' emotions, habits, experience, and interpretations of traffic conditions, as well as the varied decision-making behaviors of AVs due to manufacturers' distinct design philosophies, safety standards, and algorithmic parameters [7]. For example, some AVs might be programmed to exercise caution in lane-changing, emphasizing longer safety distances, whereas others might prioritize fluid traffic flow, opting for lane changes at reduced safety margins. Such discrepancies magnify the diversity of driving styles. Additionally, the driving styles of human drivers are shaped by a range of personal factors, emotional changes at any given moment, such as impatience or fear, can significantly affect a driver's decision-making [3]. For example, in following human-driven vehicles, AVs must be capable of rapidly adjusting to these unpredictable behaviors, altering their speed and following protocols to preserve a safe following distance. Thus, the capacity of vehicles to accurately perceive and instantaneously adapt to the dynamic nature of driving styles is critical for enhancing road safety and optimizing traffic flow.

Vehicle navigation on roads primarily involves discrete lateral lane change maneuvers and continuous longitudinal speed adjustment. Existing research either focuses on handling discrete actions, considers continuous actions, or attempts to convert discrete actions into continuous ones and vice versa to simplify the problem [6], [8]. However, this method can compromise decision-making precision and may overlook the nuanced safety benefits that subtle speed variations offer. Moreover, the performance of algorithms in terms of stability and their ability to rapidly converge toward an optimal solution is pivotal for the safety and efficacy of vehicular decision-making processes. If an algorithm cannot quickly converge to an optimized strategy within a short period, it might result in uncertain actions during lane-changing overtaking decisions, increasing the risk of traffic accidents, especially under high-speed or complex road conditions. A stable algorithm could ensure consistent decisions in complex scenarios like MCCV, while a quickly converging algorithm enables the autonomous driving system to swiftly adapt to environmental changes and timely update its decision-making strategies to accommodate new driving conditions [9]. Therefore, efficiently handling the hybrid action space in lane-changing overtaking decisions and improving the algorithm's stability and training convergence speed can further enhance the safety and efficiency of executing lane-changing overtaking maneuvers.

Considering complex interactions involved in the coexistence of multiple connected and non-connected vehicles, this study presents an overtaking strategy engineered to facilitate collaborative control over multiple connected vehicles, simultaneously prioritizing safety and augmenting the efficiency of overtaking maneuvers. First, we implement collaborative lane-changing overtaking decisions among multiple connected vehicles and propose a multi-criteria priority detection and event-triggered mechanism to optimize collaboration between vehicles. Then, acknowledging the variance in driving styles between autonomous and human-driven vehicles, we develop an algorithm to dynamically capture vehicle driving styles, significantly improving decision-making safety. Moreover, to further enhance the efficiency of collaborative lane-changing overtaking decisions among connected vehicles and effectively handle the hybrid action space involved in the decision-making process, we propose a parameterized action space algorithm based on a multi-agent paradigm. This algorithm also incorporates contrastive learning to stabilize the strategy update process and improve model stability in complex scenarios. The main contributions of this paper are as follows.

- Addressing the MCCV scenario of mixed decision-making on future roads, this paper successfully implements collaborative lane-changing overtaking decisions among multiple connected vehicles. A novel priority detection method is designed by combining criteria such as speed difference and safety conditions, and a dynamic event-triggered mechanism based on confidence intervals is proposed. This method not only facilitates real-time decision support but also effectively addresses potential competition and conflicts among connected vehicles.
- To address safety risks posed by diverse driving styles in MCCV environments, we propose an Improved Dynamic Precise Fuzzy C-Means (IDP-FCM) algorithm that integrates outlier handling, optimized cluster center initialization, and online learning to dynamically capture the driving styles of vehicles. This method significantly enhances decision-making safety, helping to mitigate potential safety hazards associated with different driving styles.

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- For the hybrid action space of multiple connected vehicles, we propose a Multi-agent Contrastive Parameterized Dueling Deep Q-Network (MCPDDQN) algorithmic. To tackle the challenges of unstable training and the need for agent-environment adaptation in distributed multi-agent reinforcement learning, we ingeniously integrate contrastive learning into the MCPDDQN algorithm. By constructing positive and negative samples from randomly sampled data in the experience pool and calculating the contrastive loss, this method not only stabilizes the strategy updates but also reduces the risk of the model making unreasonable actions, thereby enhancing the model's stability across varied scenarios.

The rest of the paper is organized as follows. Section II reviews related work. Section III describes the system model and problem. Section IV outlines our strategy. Section V discusses experimental results. Section VI concludes the paper.

II. Related Work

This section primarily reviews studies related to driving style clustering and lane changing and overtaking of vehicles.

A. Driving Style Clustering of Vehicle

In unsupervised learning tasks, clustering algorithms are instrumental in classifying data based on item similarities, playing a critical role in the analysis of vehicle data for the extraction of pivotal information regarding vehicle motion and behavior [10]. To categorize data effectively, hard clustering algorithms such as k-means and DB-SCAN are employed, adhering to the principle of minimizing intra-cluster distances, and rigidly assigning data points to a single cluster [11]. Li et al. [12] leverage principal component analysis to streamline the dimensionality of driving style features, followed by categorizing drivers into aggressive, normal, and cautious types through the k-means clustering algorithm. Mohammadnazar et al. [7] employ k-means and k-medoids methods for clustering driving style into aggressive, normal, and calm categories. This approach utilizes location-based Big Data and vehicle kinematics, such as speeds and accelerations, to differentiate driving styles. Nevertheless, in scenarios where data boundaries are blurred or overlapping, hard clustering fails to represent the possibility of data points belonging to multiple clusters. To overcome this limitation, soft clustering methods like Fuzzy C-means (FCM) allow for partial membership to multiple clusters by minimizing the overall sum of weighted intra-cluster distances. Qi et al. [13] utilize the kernel FCM clustering method to preprocess driving behavior data, categorizing it into distinct driving styles based on vectors of time gap, speed difference, and acceleration.

B. Lane Changing and Overtaking

Lane changing and overtaking are critical functionalities within autonomous driving systems [4]. Existing research focuses on utilizing advanced perception systems, decision-making algorithms, and control strategies for autonomous lane changing and overtaking, aiming to enhance the safety and driving efficiency of vehicles in intricate road scenarios [14]. He et al. [5] employ constrained observation robust Markov decision processes alongside black-box attack methods grounded in Bayesian optimization to guarantee the safety of autonomous decision-making in vehicles confronted with severe observational disturbances. This research models vehicle behavior through discrete lane-changing maneuvers. Although these studies enhance the autonomy of individual vehicles, most do not delve into the interactions and collaborative potential between vehicles. Given this, recent studies have shifted focus towards collaboration among multiple vehicles [15]. In the context of connected autonomous driving, the independent actions of single vehicles are insufficient to tackle high-density traffic and complex traffic conditions [16]. Promoting communication and collaboration among vehicles could markedly enhance the efficiency and safety of the overall traffic flow [6]. Yu et al. [17] introduce a Multi-Agent Reinforcement Learning (MARL) strategy utilizing dynamic coordination graphs, tackling the continuously changing dependencies in distributed coordination problems and providing an effective learning framework for collaborative behavior of vehicles on highways, with vehicle movements modeled as discrete following speeds.

C. Summary

In reviewing the existing research, we could identify some deficiencies concerning lane-changing overtaking. First, the bulk of research focuses on refining decision-making efficacy for CAVs during lane-changing overtaking maneuvers, frequently neglecting the impact of non-connected vehicles on the decision-making processes of connected vehicles in MCCV scenarios. Second, there is a lack of in-depth exploration of interactions between autonomous and human-driven vehicles due to differences in driving styles in the MCCV scenario, and how to capture and adapt to these style changes in real-time. Existing clustering methods for driving styles often fail to consider the dynamic and interactive nature of MCCV scenarios, potentially leading to suboptimal decisions during real-time lane-changing overtaking operations. Third, concerning the intricate action space associated with lane-changing overtaking processes, existing methods tend to simplify the treatment of hybrid action spaces. This is often achieved through the discretization of continuous actions or the transformation of discrete actions into continuous representations, which diminishes the precision in handling hybrid action spaces. Furthermore, the development of methods that enhance the stability and training efficacy of algorithms is critically important. In light of these deficiencies, this study proposes a collaborative method among multiple agents that integrates driving styles, aimed at optimizing collaborative driving among connected vehicles in the MCCV scenario.

III. System Model and Problem Formulation

This section initially presents a systematic overview of the proposed research problem. Following this, we provide detailed explanations of the relevant models, including node, statistical analysis, and safety assessment. Finally, we mathematically

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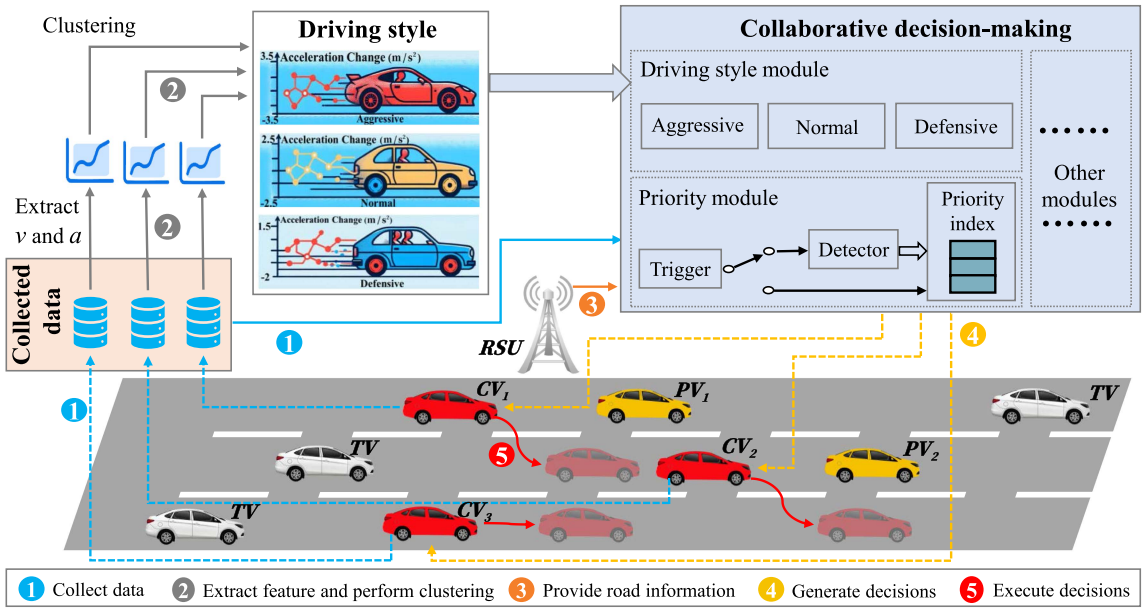

Fig. 1. System model. CV denotes connected vehicle. PV represents preceding vehicle to be overtaken. TV signifies traditional vehicle.

TABLE I
KEY NOTATIONS LIST

Notation	Description
CV, PV, TV	Connected vehicle, preceding vehicle, traditional vehicle
L, R	The set of lanes and RSUs
$relState$	The surrounding environment of the vehicle
ttc_t, ttc^{safe}	The collision time at time t, the safe collision time
$driE$, $driS$	Indicators of decision efficiency and decision safe
M, C	The set of clustered data points and cluster centers
μ	The degree of membership to three driving styles
opi_t^i	The overtaking priority index of the CV_i at time t
$d_t^i, c_{d,t}^i$	Discrete and continuous action of CV_i at time t
d^{*i}, c_d^{*i}	Optimal discrete and continuous actions of CV_i

formulate the entire research problem. Furthermore, the key notations involved in this study are shown in Table I.

A. System Overview

This paper focuses on the collaborative control of multiple connected vehicles to complete lane-changing overtaking maneuvers on highways. The overall framework is shown in Fig. 1. In a typical highway featuring three lanes, eight vehicles are selected for analysis, including three connected vehicles CVs that need to collaborate on lane-changing overtaking decisions, two preceding vehicles PVs, which refer to traditional vehicles that need to be overtaken, with the remainder being non-connected traditional vehicles TVs.

CVs drive on the road, perceiving state information of surrounding vehicles via onboard sensors. At a given time t, three CVs are required to complete a collaborative lane-changing overtaking task in an MCCV scenario. Initially, speed and acceleration data is extracted from the historical state data collected by CVs. Subsequently, these data are clustered into three distinct driving styles: aggressive, normal, and defensive. These driving styles are utilized to assess the impact of surrounding vehicles' driving behaviors on collision time (as detailed in Section IV-A). The input to the collaborative decision-making module also encompasses data on surrounding vehicles, and road information provided by the Road Side Unit (RSU). The road information primarily serves to establish thresholds for priority trigger conditions on various road segments. In the collaborative decision-making process for lane-changing overtaking, potential conflicts or competition for lane changes among CVs may arise. By integrating the current state of CVs with the thresholds for different road segments, it can be determined whether the priority detector is triggered. Once activated, the detector calculates the priority index for overtaking for all CVs within the specified range that requires lane-changing overtaking. The priorities for executing decisions among CVs are determined based on these indexes(as discussed in Section IV-B). Notably, once a priority detector is activated, the CV executing the overtaking continues to utilize the currently calculated priority index until the overtaking maneuver is completed. Subsequently, this priority index is no longer in use. Ultimately, the lane-changing overtaking strategy for each CV is generated through a collective analysis incorporating the driving style module, the priority module, and other modules (described in Section IV-C). Based on these driving strategies, CVs execute the designated maneuvers.

B. Node Model in MCCV Scenario

In consideration of the coexistence of connected vehicles and non-connected traditional vehicles on the road, they are denoted by the sets CV and TV, respectively. In this study, we focus on the critical information within the vehicle state data regarding position (x, y), speed v, and acceleration a. Notably, TV is incapable of data communication with CV and RSU. Given the sensing capabilities of multiple sensors equipped on vehicles, the effective sensing range of the CV to surrounding vehicles

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is set to no more than 150 m [4]. To more accurately simulate human driving behavior, the Minimizing Overall Braking Induced by Lane Change (MOBIL) model is employed for lateral lane-changing decisions of TV [18]. Concurrently, to simulate longitudinal speed control of TV, the Intelligent Driver Model (IDM) is utilized [4].

As illustrated in Fig. 1, the vehicle movement direction is assumed from left to right, and the set of lanes is denoted as L. Within this direction, the rightmost lane is represented as l_1, while the leftmost lane is represented as l_n. The RSUs along the road are represented as $R = \{R_1, R_2, \ldots, R_n\}$. To ensure the communication and storage capabilities of RSUs, each RSU is usually equipped with a small base station, responsible for performing computation and data storage for related services [19]. In this work, RSUs primarily collect and store state data of all vehicles within their coverage area over a period of time through their deployed sensor equipment and data sharing with CVs [19].

C. Analytical Model of Vehicle Surroundings

To capture the state information of other vehicles around the target vehicle more accurately, we partition the surrounding environment of the target vehicle into six different regions: left preceding, left rear, direct preceding, direct rear, right preceding, and right rear, represented by $relState = \{LP, LR, DP, DR, RP, RR\}$. These regions are defined relative to a coordinate system established with the current position of CV at the center. We focus on the current lane and the two adjacent lanes as the area of interest, taking them as the x-axis direction, while the driving direction is considered the y-axis direction. Considering the assumed perception distance of 150 m for CVs, a range of 150 m is selected both in front of and behind the CV along the y-axis to determine six regions. In each region, we adopt a strategy of selecting only the vehicle closest to CV in the direction of travel. When the CV is located at the edges (i.e., l_1 or l_n), the system considers the lack of a lane on one side and adjusts the regions accordingly. For example, if the CV is in l_n, the LP and LR regions are set to -1 to indicate the absence of vehicles on that side. Furthermore, if a region lacks any vehicles, its corresponding entry in $relState$ is set to -1. It is important to note that in scenarios where an additional lane, such as a service lane, is adjacent to l_1 or l_n, the system extends its area of interest to include the additional lane. For instance, if the CV is in l_1 and there is a service lane to the right, the RP and RR regions will include vehicles from the service lane.

D. Statistical Analysis Model of Overtaking Index

A statistical analysis model is designed to process and analyze the data of vehicles executing lane-changing overtaking maneuvers. The primary goal of this model is to accurately capture and understand the triggering conditions for CVs to perform lane-changing overtaking maneuvers.

Specifically, we first extract two pivotal metrics from the data collected by the RSU: the relative distance and the relative speed to the vehicle ahead at the onset of a lane-change overtaking action. Subsequently, the Z-score method (i.e., $Z = \frac{X-\mu'}{\sigma}$) is employed for data filtering, where X is the data point, μ' and σ denote the mean and standard deviation of the dataset, respectively [20]. We retain data points where $|Z|$ is less than a predetermined threshold to minimize outliers. Following this, the filtered data X' is transformed logarithmically ($Y = \log(X')$) to stabilize variance and approximate a normal distribution more closely. Based on Y, the mean $\bar{Y}$ and standard deviation σ_Y can be calculated. We further compute the standard error $SE = \frac{\sigma_Y}{\sqrt{n}}$, where n is the number of effective data points. Finally, $\bar{Y}$, SE and (1) are utilized to calculate the 95% confidence interval.

$$\bar{Y} \pm Z_{\alpha/2} \cdot SE \tag{1}$$

where $Z_{\alpha/2}$ represents the critical value of 1.96 for a 95% confidence level [21].

E. Basic Safe Estimation Model

Time to Collision (TTC) is widely utilized as a metric to assess the potential collision time between a vehicle and surrounding obstacles or other vehicles [14]. In RL, an agent selects and executes actions based on the current state, and the safety of the executed actions can be evaluated by calculating the TTC between two vehicles, the calculation is as follows:

$$ttc_t = \min\left(\frac{d_{pre} - d_{ego}}{v_{ego} - v_{pre}}, \frac{d_{ego} - d_{rear}}{v_{rear} - v_{ego}}\right) \tag{2}$$

where d_{ego}, d_{pre}, and d_{rear} respectively signify the positions of the ego vehicle and the vehicles preceding and behind it, while v_{ego}, v_{pre}, and v_{rear} represent the speeds of ego vehicle and vehicles preceding and behind it. The actions taken by CV at each time interval not only affect vehicles in the preceding of the target lane but also have an impact on the vehicles behind the target lane. Consequently, it is necessary to focus on the shortest possible time for a potential collision. In our study, we define a safe collision time ttc^{safe} as 3 s.

F. Problem Formulation

This paper is dedicated to exploring the collaborative optimization of lane-changing overtaking behaviors among multiple CVs, to minimize the time required for lane-changing overtaking while ensuring safety.

a) Decision Efficiency: The efficiency of lane-changing overtaking is pivotal for enhancing road capacity and reducing travel time. Elevating the velocity of vehicles during lane-changing overtaking decision-making phases can curtail the cumulative time requisite for executing such maneuvers. To this end, we model the decision-making process as a sequence of equal-temporal intervals, denoted by $t = 0, 1, 2, \ldots, T$. Decision efficiency can be represented by (3).

$$driE = \sum_{t=0}^{T} \sum_{i=1}^{N} v_{i,t} \tag{3}$$

where $v_{i,t}$ represents the speed of CV_i at time t, and N is the total number of vehicles participating in collaborative control.

b) Decision Safety: The paramount importance of maintaining vehicle safety during lane-changing overtaking maneuvers

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necessitates minimizing collision risks and enhancing collaboration among vehicles. The quantification formula for safety decisions is as follows:

$$driS = \sum_{t=0}^{T} \sum_{i=1}^{N} ttc_{i,t} \tag{4}$$

where $ttc_{i,t}$ represents the estimated time to collision for CV_i with surrounding vehicles at time t, the calculation of ttc detailed in (2).

Overall, we propose an objective function that seeks to maximize the aggregate of decision efficiency and safety:

$$\max_{v_{i,t}, ttc_{i,t},} \quad \alpha \cdot driE + \beta \cdot driS \tag{5a}$$

$$\text{s.t.} \quad C_1 : v_{i,t} \in [v_{min}, v_{max}], \forall i, \forall t \tag{5b}$$

$$C_2 : ttc_{i,t} \geq ttc^{safe}, \forall i, \forall t. \tag{5c}$$

where α and β represent weight coefficients that balance the $driE$ and $driS$. Constraint C_1 stipulates that the v for any CV_i at any moment t must be constrained within predefined speed bounds. Constraint C_2 ensures the TTC for any CV_i at any given time t exceeds a predefined ttc^{safe}.

IV. Proposed Strategy

In this section, considering the impact of variations in driving styles on lane-changing overtaking maneuvers, we propose an IDP-FCM algorithm to dynamically identify driving styles. Aiming to mitigate potential competitive conflicts between CVs during lane-changing overtaking, we develop a priority detection and dynamic event-triggered method. This is complemented by the introduction of a MARL strategy, designed to facilitate collaborative lane-changing overtaking among CVs. To validate the effectiveness of our proposed algorithm, the final part of this section analyzes the algorithm's time complexity.

A. IDP-FCM Algorithm

To alleviate the inconsistency of algorithms for AVs and the impact of personalized driving styles of human-driven vehicles on lane-changing overtaking in the MCCV environment, this study proposes an IDP-FCM algorithm for identifying vehicle's driving styles. This algorithm comprises the following four main components: employing the FCM algorithm as the foundational component, handling outliers using the Interquartile Range (IQR), initializing cluster centers using the k-means algorithm, and adapting to dynamic data through online learning techniques.

c) FCM: The identification of driving behaviors and styles hinges on key parameters like vehicle speed and acceleration. The data vector for clustering is defined as $M = \{m_1, m_2, \ldots, m_n\}$, where $m = (v_{i,t}, a_{i,t})$ corresponds to a data point composed of the speed and acceleration of ith vehicle at time t. These data is subsequently segmented into c clusters, denoted as $G = \{g_1, g_2, \ldots, g_c\}$, where $c \ll n$, and the cluster centers are denoted as $C = \{c_1, c_2, \ldots, c_c\}$.

Given the continuity and fuzziness of vehicle's driving styles, traditional hard clustering methods struggle to capture this relationship. To bridge this gap, we adopt the FCM algorithm, which partitions M into G with c cluster centers and generates a membership matrix U. This matrix quantifies the fuzziness level of each m to each g. Our goal is to minimize the distance between m and its belonging g, while iteratively adjusting U and C to find the optimal clustering result. The objective function J is expressed as follows:

$$J(U, C) = \sum_{i=1}^{n} \sum_{j=1}^{c} \mu_{i,j}^{\kappa} \left\| m_i - c_j \right\|^2, s.t. \sum_{j=1}^{c} \mu_{i,j} = 1, \mu_{i,j} \geq 0 \tag{6}$$

where $\mu_{i,j}$ denotes the membership degree of m_i to c_j and κ is the fuzziness index that controls the fuzziness degree of the membership. Finally, U and V are iteratively updated until convergence using the following equations:

$$\mu_{i,j}^{(t+1)} = \frac{1}{\sum_{k=1}^{c} \left(\frac{\|m_i - c_j^{(t)}\|}{\|m_i - c_k^{(t)}\|} \right)^{\frac{2}{\kappa-1}}} \tag{7}$$

$$c_j^{(t+1)} = \frac{\sum_{i=1}^{n} \left(\mu_{i,j}^{(t+1)} \right)^{\kappa} m_i}{\sum_{i=1}^{n} \left(\mu_{i,j}^{(t+1)} \right)^{\kappa}} \tag{8}$$

where $\mu_{i,j}^{(t+1)}$ indicates the membership degree of m_i to c_j at iteration $t+1$, and $c_j^{(t+1)}$ represents the position of the jth cluster center at iteration $t+1$, with t denoting the iteration number.

d) IQR: The presence of outliers in calculating the centroid of the cluster can lead to inaccurate positioning of the cluster centers, affecting the overall quality of clustering. To mitigate this issue, we introduce IQR into the FCM algorithm for effective outlier detection and removal [22].

Prior to the computation of U and the update of C in the FCM clustering process, we calculate the first ($Q1$) and third ($Q3$) quartiles for the v and a within M. The IQR is then computed using $IQR = Q3 - Q1$. Following this, the boundary for identifying outliers is established as (9), where k is set to 1.5. This boundary helps filter out outliers, thereby reducing noise and ensuring more accurate and reliable clustering by the FCM algorithm.

$$\text{AnaRan} = \{x \mid x < Q1 - k \times IQR \,\text{or}\, x > Q3 + k \times IQR\} \tag{9}$$

e) Initial Cluster Centers: The selection of initial cluster centers impacts the clustering results of the FCM algorithm. To optimize this phase, we integrate the k-means algorithm to determine initial cluster centers. The process includes the following steps: Initially, c data points are randomly selected as initial cluster centers. Subsequently, the following sub-steps are iterated until the cluster centers stabilize:

- Each m is assigned to its nearest cluster centers, thereby forming c distinct clusters denoted as g.
- The mean of each g is computed and designated as the new cluster centers.

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These stabilized cluster centers serve as the starting point for the FCM algorithm, ensuring the algorithm begins its iterations from a reasonable position, thereby enhancing both the convergence speed and the clustering result's accuracy.

f) Online Clustering: Considering that the process of entire lane-changing overtaking unfolds over several seconds or more, we apply the improved FCM algorithm for online clustering to handle continuously arriving data. Specifically, after obtaining the clustering result at time t, as the overtaking time progresses, our approach adapts to dynamic data in two primary steps. First, for each new data point m_{new}, its membership degree $\mu_{new,j}$ is calculated as shown in (7). Subsequently, the incorporation of m_{new} into the M causes the position of C to shift. This can be adjusted using (10).

$$c'_j = (1 - \mu_{new,j}) \cdot c_j + \mu_{new,j} \cdot m_{new} \tag{10}$$

where c'_j represents the updated cluster center, and $\mu_{new,j}$ is used to control the impact of the new data point on the cluster center update. This online clustering approach not only allows the algorithm to adjust quickly to new information but also preserves the clustering information of existing data, reducing the need for frequent recalculations of the entire dataset.

g) Driving Style Determination Criteria: The driving styles are categorized into three categories: aggressive, normal, and defensive [10]. To correlate the clustering results with these categories, we design the following criteria:

- Average speed and acceleration ($\overline{v}$, $\overline{a}$): For each cluster, we compute $\overline{v}$ and $\overline{a}$, which directly reflect the basic characteristics of driving behavior. Typically, an aggressive driving style is associated with higher $\overline{v}$ and $\overline{a}$.
- Standard deviation of speed and acceleration (σ_v, σ_a): To measure the consistency and variability of driving behavior within each cluster, we calculate σ_v and σ_a. High σ indicates diversity and instability in driving behavior within the cluster, a characteristic possibly of aggressive driving styles.

Composite Score ($score$): To holistically consider these criteria, we calculate $score_g$ for the gth cluster using (11).

$$score_g = w_1 \cdot \overline{v_g} + w_2 \cdot \overline{a_g} + w_3 \cdot \sigma_{v,g} + w_4 \cdot \sigma_{a,g} \tag{11}$$

where w_1, w_2, w_3, and w_4 are the weights for $\overline{v}$, $\overline{a}$, σ_v, and σ_a, respectively. The cluster with the highest $score$ is classified as aggressive, the one with the lowest $score$ as defensive, and the other cluster as normal.

h) IDP-FCM: CVs navigating on the road continuously collect state information of surrounding vehicles using onboard sensors. To ensure the relevance and timeliness of the vehicle driving data, we construct M using historical data collected by CV over the previous 6 s. Upon CV initiates lane-changing overtaking maneuvers, the IDP-FCM algorithm is activated to cluster v and a generated by the vehicle and its surrounding vehicles, determining the membership degrees of surrounding vehicles to various driving styles. To mitigate the impact of erroneous data, which may arise from emergency braking or sensor inaccuracies on the clustering results, we utilize the IQR method to filter out these outliers. Additionally, we adopt cluster centers derived from the k-means algorithm's clustering as the initial cluster centers for FCM to boost the convergence speed and stability of the algorithm. Given that the completion of lane-changing overtaking actions may span a considerable duration, during which the vehicle's driving state and behavior could undergo multiple changes, we incorporate an online learning technique to perform dynamic online clustering. The IDP-FCM algorithm concludes once CV has successfully executed the overtaking of the target vehicle. These improvements significantly enhance the algorithm's accuracy and applicability in dynamic driving scenarios, offering a novel approach for the effective identification of different driving styles and reducing the risk of lane-changing overtaking. The complete algorithm is detailed in Algorithm 1.

Algorithm 1: IDP-FCM Algorithm.

Input: Vehicle state data: velocity (v) and acceleration (a).
Output: Membership degrees μ of three driving styles.

```
CVs perceive and store the v and a of surrounding vehicles in past 6 s.
CVs determine the relState of surrounding environment.
The dataset for selected vehicles is denoted as M'.
for i in N_CV do
    if CV_i requests to lane-changing overtaking then
        Initialize overtaking time, t_i ← 0.
        while CV_i do not complete overtaking do
            if t_i = 0 then
                Apply (9) to remove outliers from M', forming M.
                Use k-means to determine optimal cluster centers C'.
                Utilize C' as initial cluster centers for FCM clustering.
            else
                Use (7) to calculate μ of new data m_new.
                Update C' online using m_new, μ and (10).
            end if
            t_i++.
        end while
    end if
end for
```

B. Priority Detection and Triggering

This subsection primarily discusses two core aspects: priority detection and dynamic event-triggered mechanisms.

1) Priority Detection: In scenarios involving multiple CVs attempts to execute lane-changing overtaking simultaneously, handling potential conflicts is essential. To address this, we develop a priority detection module that assesses the urgency and safety of each CV for overtaking based on the following key factors:

- Speed Difference (f_v): The imperative for overtaking emerges when the speed of CV significantly exceeds that of the preceding vehicle in the same lane, to avoid affecting

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the smooth flow of trailing vehicles. The f_v is expressed as $f_v = v_{ego} - v_{pre}$.

- Safety Conditions (f_s): Adequate spacing is essential for safe overtaking. As the distance to the preceding vehicle decreases, indicating a higher risk of collision, the necessity for overtaking correspondingly increases. The f_s is calculated given by:

$$f_s = -\log \frac{d_{preceding} - d_{ego} - h_v}{t_h \cdot (v_{ego} - v_{preceding})} \tag{12}$$

where t_h denotes the headway threshold, which is set to 1.2 s, and h_v represents vehicle length.

- Traffic Flow (f_t): The necessity for lane change arises when shifting to a less congested lane could enhance traffic flow. The f_t is calculated as $f_t = \frac{tr_c}{tr_a}$, where tr_c and tr_a denote the vehicle counts in the current and alternative lanes, respectively, measured over the same time period and road segment length.

By considering these factors, we develop an Overtaking Priority Index (opi) for evaluating the immediate priority of CV_i for lane-changing overtaking:

$$opi_t^i = \omega_1 f_v^i + \omega_2 f_s^i + \omega_3 f_t^i + \epsilon_t \tag{13}$$

where ω_1, ω_2, and ω_3 are the weighting coefficients for the respective factors. ϵ_t is a small random variable obeying $\mathcal{N}(0, 0.001)$, ensuring diversity in priority index.

2) Priority Trigger: Determining the optimal timing for priority detection is critical. This study proposes a priority triggering module based on confidence intervals for efficient decision-making and computational load reduction. RSUs collect and analyze vehicle state data within a specific time window of the covered road sections, focusing on two critical attributes as vehicles initiate lane-changing overtaking maneuvers: relative distance and relative speed to the preceding vehicle. Leveraging the statistical analysis model delineated in Section III-D, we calculate the 95% confidence intervals for these attributes. Considering different road sections may exhibit different driving conditions, the confidence intervals calculated by each RSU may differ, hence vehicles need to determine whether to trigger priority detection based on the current road section information. Detailed experimental validation is provided in Section V-B1. Furthermore, for real-time decision support, we devise a criterion for triggering judgments for CV_i at time t, enabling the real-time surveillance of data emanating from all CVs. Each new data point X_{new} is transformed using the logarithm function $\log(X_{\text{new}})$ to align with the statistical processing treatment employed. If CV_i falls within the coverage range of R_j at time t, the conditional judgment is as follows:

$$\eta_i(t) = \begin{cases} 1, \text{if } \log(X_{\text{new}}) \in [\bar{Y}_j - Z_{\alpha/2} \cdot SE_j, \\ \qquad \bar{Y}_j + Z_{\alpha/2} \cdot SE_j] \\ 0, \text{otherwise} \end{cases} \tag{14}$$

Considering the imperatives of real-time performance and computational efficiency, we design a dynamic event-trigger mechanism for CV_i:

$$t_i^0 = 0, t_i^{k+1} := \inf\left\{t \geq t_i^k + \zeta_{\min} | \eta_i(t) = 1\right\} \tag{15}$$

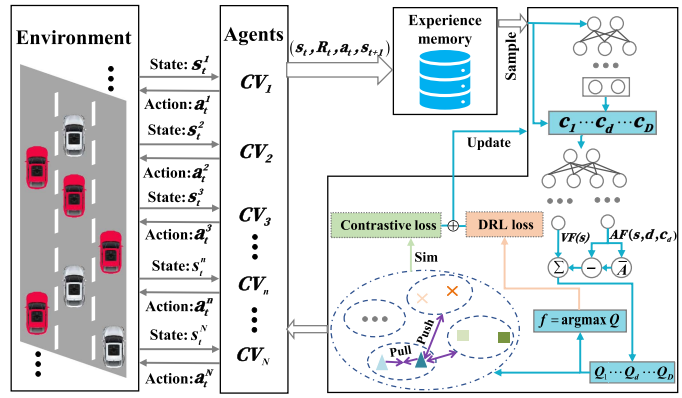

Fig. 2. Framework of MCPDDQN algorithm.

where $\zeta_{\min}$ signifies the minimum interval required for each operational execution. This mechanism effectively mitigates computational load on CVs, thereby augmenting the real-time precision and efficiency of inter-vehicle communication and ensuring robust decision support in dynamic traffic situations.

C. Lane-Changing Overtaking Strategy MARL-Based

This subsection introduces how DRL is used to control CVs in MCCV environments to execute lane-changing overtaking maneuvers. The algorithm structure is depicted in Fig. 2.

1) Preliminary: Addressing the issue of lane-changing overtaking decisions by CVs in MCCV environments, we model the maneuver decision-making process of CVs as a Parametric Action Markov Decision Process (PAMDP). In this modeling, the state space S not only reflects the current state of the vehicle but also dynamically incorporates the surrounding environment and its interactions with other vehicles. The action space A is parameterized to include various maneuver actions the vehicle might undertake. The state transition probability $\mathcal{T}$ describes the likelihood of transitioning from the current state to future states, while the reward function $\mathcal{R}$ assesses the value of actions based on the current state. Moreover, the discount factor $\gamma \in [0, 1]$ is used to balance short-term and long-term rewards in decision-making. For a comparative analysis of γ, please see Section I-A1 in the supplementary file. Consequently, our PAMDP model can be represented as $P = (S, A, \mathcal{T}, \mathcal{R}, \gamma)$.

To mitigate uncertainty issues brought by independent learning, our algorithm employs centralized training with a decentralized execution strategy within the MARL paradigm [6], [23]. Specifically, each agent (i.e., CV) independently records its observed state and executes actions output by the shared policy. The experiences gathered from all CVs are then aggregated to refine the policy governing lane-changing overtaking decisions.

2) State Space: We meticulously define the state space to include the state of the CV itself as well as the relative state of surrounding vehicles. The state of a CV primarily includes its v, a, lane identification number (l), and the distance from the start of the current road segment (d). These parameters provide dynamic characteristics and location information of the CV, which are crucial for the decision-making process.

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Furthermore, state space is extended to incorporate the state of nearby vehicles. Based on $relState$ outlined in Section III-C, we focus on the state of vehicles within six defined regions around the CV, including their speeds (v_j) and longitudinal distance differences ($\triangle d_{j,i}$) relative to CV. This method not only captures the immediate traffic dynamics around the CV but also aids in understanding the interactive behaviors within the traffic flow. Consequently, complete state of CV_i at each time slot is defined as $S_i = \{v_i, a_i, l_i, d_i, \{v_j, \triangle d_{j,i}\}_{j=0}^{5}\}$. In cases where specific areas within $relState$ lack the presence of vehicles, the corresponding v_j and $\triangle d_{j,i}$ are set to -1.

3) Action Space: The decision-making process for lane-changing overtaking by CVs involves a breakdown of action space into two main types: lateral lane-changing and longitudinal speed control [8]. The lateral lane-changing actions are discrete and are intended to determine whether the vehicle needs to change lanes and in which direction, denoted as $A_l \in \{-1, 0, 1\}$, where -1 indicates a lane change to the left, 0 represents staying in the current lane, and 1 signifies a lane change to the right. Additionally, longitudinal speed control A_v is designed as a continuous action set. Considering requirements of traffic regulations and vehicle characteristics, the range of A_v is defined as $[0,\ 30]$m/s.

4) Reward Function: To optimize the safety and efficiency of CVs, we design a multifaceted reward function that integrates the following five factors: the safety index incorporating driving styles, speed execution efficiency, driving harmony, decision priority, and penalties for unreasonable actions.

a) Safety Index Incorporating Driving Style (r_{sds}): In the MCCV scenario involving autonomous and human-driven vehicles, the diversification of driving styles due to different safety standards and driving habits leads to varying safety risks. We calculate the r_{sds} using the formula:

$$r_{sds} = ttc \cdot \sum_{i=1}^{3} (\Theta_i \cdot \mu_i) \tag{16}$$

where $\Theta \in (-0.5, 0.5)$, which denotes impact coefficients for three distinct driving styles (aggressive, normal, defensive). For the specific value of Θ, please refer to Section I-A2 in the supplementary file. The μ represents the membership degrees to these driving styles. The ttc is detailed in Section III-E.

b) Speed Execution Efficiency (r_{ve}): This component is bifurcated into two sub-parts: comfort level of the vehicle's speed changes (r_{comf}) and the relative speed difference (r_v), specific calculations are shown in (17) and (18). The former incentivizes smooth driving by penalizing abrupt speed changes, while the latter encourages CVs to minimize the speed gap with the vehicle preceding, facilitating quicker journey speed. So, r_{ve} is expressed as $r_{ve} = r_{comf} + r_v$.

$$r_{comf} = -\frac{|a| - a_{\min}}{a_{\max} - a_{\min}} \tag{17}$$

$$r_v = \frac{v - v_{preceding} - v_{\min}}{v_{\max} - v_{\min}} \tag{18}$$

c) Driving Harmony (r_{dh}): The r_{dh} is used to evaluate the effect of lane-changing actions by CV on the traffic flow within the target lane, it is calculated using (19). It measures the relative improvement or deterioration in the speed of following vehicles in the target lane post-maneuver. A positive r_{dh} indicates an improvement in traffic flow, whereas a negative value suggests a decrease in traffic efficiency due to the maneuver.

$$r_{dh} = \frac{v_{after}^{follow} - v_{before}^{follow}}{v_{\max}} \tag{19}$$

d) Decision Priority (r_{dp}): To promote cooperative driving among CVs, we calculate a priority score for each CV engaged in lane-changing overtaking maneuvers. This score takes into account lane yielding (r_{GL}) and speed yielding (r_{GV}), which assists high-priority CVs in making decisions with priority. The collaborative reward of CV_i is defined as:

$$r_y = \begin{cases} (opi_j - opi_i) \cdot opi_i, & \text{if } CV_i \text{ yields lane or speed} \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

$$r_{dp} = \alpha_1 \cdot r_{GL} + \alpha_2 \cdot r_{GV} \tag{21}$$

where α_1 and α_2 are weights, j denotes the index of the CV involved in the collaborative maneuver, and opi is calculated according to (13). Both r_{Gl} and r_{GV} are calculated by r_y.

e) Penalty for Unreasonable Actions (r_{no}): We penalize CVs that produce unexecutable or unsafe actions, such as making left lane changes on the far-left lane and making unnecessary lane changes when there is no vehicle ahead. Additionally, CVs that fail to overtake the target vehicle ahead are also penalized. The penalty is set at -1 for any action meeting these conditions.

The total reward function for CV_i, integrating all the aforementioned components, is expressed as:

$$r^i = w_1 \cdot r_{sds}^i + w_2 \cdot r_{ve}^i + w_3 \cdot r_{dh}^i + w_4 \cdot r_{dp}^i + w_5 \cdot r_{no}^i \tag{22}$$

where w_1, w_2, w_3, w_4, and w_5 are weights of the above five components respectively. To validate the positive impact of specific modules on overall performance, we conduct a series of ablation experiments, with the experimental results detailed in Section I-B in the supplementary file. For the weight ratios between various factors, please refer to Section I-A3 in the supplementary file, which contains a detailed analysis of the impact of different weight ratios on overall performance.

5) MCPDDQN Algorithm: To tackle the intricate task of collaborating multiple CVs to execute lane-changing overtaking maneuvers, we propose an MCPDDQN algorithm that not only controls multiple CVs to perform lane-changing overtaking actions but also effectively handles the discrete lateral lane-changing actions and continuous longitudinal speed actions. The algorithm is built upon an Actor-Critic framework, where the actor-network utilizes Proximal Policy Optimization (PPO), and the critic network employs a Dueling Deep Q-Network (DDQN). The essence of this algorithm lies in the parametrization of the action space, linking each discrete action with an associated continuous action, and employing a hierarchical approach to ascertain the optimal discrete and its corresponding continuous actions.

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For any action a_t^i taken by CV_i at time t, it is parameterized into a pair of discrete action $d_t^i \in D^i$ and corresponding continuous action $c_{d,t}^i \in C_d^i$. To effectively evaluate the effect of CV_i performing a_t^i, we define a state-action value function $Q(s^i, a^i) = Q(s^i, d^i, c_d^i)$ to quantify the utility of performing action a under the current state s. The relationship between $Q(s*, *a)$ for CV_i performing a under s and the expected return from its subsequent state-action pairs is represented by constructing the Bellman equation, as specified in the following:

$$\begin{aligned} &Q\left(s_t^i, d_t^i, c_{d,t}^i\right) \\ &= \mathop{\mathbb{E}}_{r_t^i, s_{t+1}^i}\left[r_t^i + \gamma \max_{d_t^i \in D^i} \sup_{c_{d,t}^i \in C_d^i} Q\left(s_{t+1}^i, d_t^i, c_{d,t}^i\right)\right. \\ &\left. \left| s_t^i = s^i, a_t^i = \left(d_t^i, c_{d,t}^i\right)\right.\right] \end{aligned} \tag{23}$$

Given that multiple viable continuous action solutions might exist for a specific discrete action, the PPO algorithm is utilized to map the state s^i of CV_i to the optimal continuous action c_d^{*i}, defined as $c_d^{*i} = \pi(s^i; \theta_\pi^i) = \text{argmax}_{c_d^i \in C_d^i} Q(s^i, d^i, c_d^i)$, where $\pi(\cdot)$ represents the policy function to obtain the best c_d from C_d, θ_π^i denotes the set of parameters for solving the optimal action under parameters θ_π for s^i when training the PPO network.

After generating c_d, the critic network constituted by the DDQN algorithm parameterizes the value function Q by decomposing the DDQN action value function $Q(s, a)$ into a state value function VF and an action advantage function AF. This decomposition facilitates the determine of the optimal discrete action d^{*i} for CV_i as follows:

$$\begin{cases} Q(s^i, d^i, c_d^i; \theta_Q^i) = VF(s^i; \theta_{VF}^i) + AF(s^i, d^i, c_d^i; \theta_{AF}^i) \\ d^{*i} = \underset{d^i \in D^i}{\text{argmax}} Q(s^i, d^i, c_d^i) \end{cases} \tag{24}$$

where θ_Q, θ_{VF}, and θ_{AF} respectively denote the training parameters of DDQN, the parameter sets for the VF and the AF. Through this methodology, we can effectively determine the optimal d and its corresponding c_d.

The MCPDDQN algorithm utilizes the Temporal Difference (TD) method for parameter updates, executing this process in two distinct phases. Initially, parameters of the Q-value network, θ_Q, remain constant, whilst θ_π of the PPO policy network undergo optimization by minimizing the TD loss. Subsequently, both policy network parameters θ_π and value function network θ_Q in the DDQN are updated, primarily through the negative gradient of the quality evaluation of d. The method of loss calculation for CV_i is as shown in (25).

$$\begin{cases} y_t^i = r^i + \gamma \max_{d^{i'} \in D^i} Q\left(s^{i'}, d^{i'}, \pi\left(s'; \theta_\pi\right); \theta_Q\right) \\ \mathcal{L}_Q^i\left(\theta_Q\right) = \frac{1}{2}\left[Q\left(s^i, d^i, \pi\left(s^i; \theta_\pi\right); \theta_Q\right) - y_t^i\right]^2 \\ \mathcal{L}_\pi^i\left(\theta_\pi\right) = -\sum_{d^i=1}^{D^i} Q\left(s^i, d^i, \pi\left(s^i; \theta_\pi\right); \theta_Q\right) \end{cases} \tag{25}$$

where y_t represents target state-action value, and s' denotes next state of CV. The loss functions ℓ_Q and ℓ_π correspond to DDQN and DDPG algorithms, respectively, with θ indicating parameter update policy. The algorithm employs stochastic gradient descent techniques to optimize aforementioned loss functions based on θ alterations, minimizing prediction errors to augment algorithm performance.

To enhance the model's stability and convergence speed across various road scenarios, a contrastive learning scheme is introduced [24]. This strategy optimizes the generated model strategy by discriminating between positive and negative exemplars. A novel approach is designed by comparing the action value predicted under the current policy, $\mathbf{y}_p$, with the target action value $\mathbf{y}_e$. Samples exhibiting higher Q-values in RL's experience pool are considered positive, while those with lower Q-values are deemed negative. This approach more effectively leverages sample data, enhancing training results.

In algorithm implementation, a contrastive loss $\mathcal{L}_{con}$ is employed to refine training effects. Initially, the cosine similarity among positive samples is computed [24], followed by the evaluation of the cosine similarity for the action values of negative samples, $\mathbf{y}_k$, through a comparison between the predicted action value under the current policy and the action values of negative samples. By minimizing the loss function through negative gradient optimization and incorporating the similarity data of both positive and negative samples, the equation of $\mathcal{L}_{con}$ is as follows:

$$\mathcal{L}_{con} = -\log \frac{e^{(\text{sim}(\mathbf{y}_p, \mathbf{y}_e)/\tau)}}{e^{(\text{sim}(\mathbf{y}_p, \mathbf{y}_e)/\tau)} + \sum_{k=0}^{K-1} e^{(\text{sim}(\mathbf{y}_p, \mathbf{y}_k)/\tau)}} \tag{26}$$

where the cosine similarity function $\text{sim}(\mathbf{y}_1, \mathbf{y}_2) = \frac{\mathbf{y}_1^T \mathbf{y}_2}{\|\mathbf{y}_1\|_2 \|\mathbf{y}_2\|_2}$, K denotes the size of selected batch data, τ is the temperature parameter, controlling the model's sensitivity to differences between positive and negative samples.

Ultimately, by amalgamating the contrastive loss with the Q-value estimation loss, a comprehensive loss function is formulated within the critic network, as delineated in (27), thereby considering both the action value prediction error and the contrastive learning effect to improve the algorithm's training stability and convergence speed.

$$\mathcal{L}_{total}^Q = \mathcal{L}_Q + \mathcal{L}_{con} \tag{27}$$

6) Complexity Analysis: We conduct a time complexity analysis of the proposed MCPDDQN algorithm. During the centralized training phase, the MCPDDQN algorithm leverages a fully connected neural network structure. The time complexity of this phase can be articulated as $\mathcal{O}(E \cdot (\sum_{i=1}^{L_{ppo}-1} N_i^{ppo} N_{i+1}^{ppo} + \sum_{i=1}^{L_{ddqn}-1} N_i^{ddqn} N_{i+1}^{ddqn}))$, where L_{ppo} and L_{ddqn} respectively denote the number of layers in the actor and critic networks, N_i^{ppo} and N_i^{ddqn} respectively denote the neuron count in the ith layer of the corresponding networks. E signifies the total number of episodes during the distributed learning process. It is important to highlight that this training phase can be performed offline, which significantly alleviates the pressure on real-time computation. Moving into the distributed execution phase, the time complexity is streamlined to $\mathcal{O}(\sum_{i=1}^{L_{ppo}-1} N_i^{ppo} N_{i+1}^{ppo} + \sum_{i=1}^{L_{ddqn}-1} N_i^{ddqn} N_{i+1}^{ddqn})$. In each time step within this phase, the actor and critic networks of all agents' models execute once to determine each agent's discrete action and its corresponding continuous action. With the number of hidden layers and neurons

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TABLE II
PARAMETER COMPARISON OF TEMPERATURE COEFFICIENT

τ	0.001	0.01	0.05	0.1	0.15	0.2	0.3
conS	740	940	800	1040	1680	1390	1290
conAR	3	3.2	4.2	4.1	3.7	4.1	4

fixed, the complexity of the MCPDDQN algorithm is primarily influenced by the sizes of the input and output layers. Based on the state space and action space designed in Section IV-C, the input layer dimension of the actor-network is determined to be 16, while the dimension of the output layer is 3. Simultaneously, the input layer dimension of the critic network is 19 (i.e., $16 + 3$), with an output layer dimension of 3, used to evaluate the value of a given state and action combination.

V. RESULTS AND ANALYSIS

This section presents a series of experiments conducted to validate the effectiveness of our proposed strategy. Initially, the simulation environment and configuration of related parameters are detailed. Subsequently, the feasibility of various modules within the strategy is evaluated. Then, an experimental comparison is made by integrating existing related work. Finally, the performance of our proposed strategy under different penetration rates of CV is validated.

A. Simulation Setup

The simulation is carried out on a computer configured with an Intel(R) Xeon(R) Gold 6226R CPU @2.90 GHz 2.89 GHz. The source code for our project is available at link.[1] For the construction of the simulation environment, we utilize the Simulation of Urban MObility (SUMO) tool to build a unidirectional three-lane straight road with a length of 3000 m [5]. Vehicles are randomly distributed across each lane, with separations varying from 40 m to 150 m, and three of these vehicles are selected for control by the algorithm as connected vehicles, while the remaining vehicles are set to follow the MOBIL and IDM models as non-connected vehicles. To select the optimal temperature coefficient τ for the contrastive learning involved in the study, we evaluate τ based on the convergence speed ($conS$) and the average reward at convergence ($conAR$). The evaluation results, as shown in Table II, led us to the final selection of $\tau = 0.05$. For more details on the parameters refer to Table III.

B. Results and Analysis

This subsection evaluates the performance of our proposed strategy's individual modules and validates its effectiveness through comparison experiments.

1) Threshold Selection for Different Roads: To substantiate the premise articulated in Section IV-B2 that different roads may present varying driving conditions, we conduct a verification using data on US101 and I80 roads from the Next Generation Simulation (NGSIM) dataset [25] and a simulation environment

[1][Online]. Available: https://github.com/NetworkCommunication/LCOMCCV

TABLE III
PARAMETERS SETTING

Experimental Parameter	Value
Vehicle length	5m
Lane width	3.2m
Road length	3000m
Number of lanes	3
Max lane speed	30m/s
Number of CVs	3
Initial separation distance between TVs	[40, 150]m
Time step for each vehicle movement	0.1s
Total number of vehicles	30
Maximum time step per episode	230
Batch size	128
Experience pool size	5000
Learning rates of actor network and critic network	0.00001, 0.00001
Total training episodes	2000

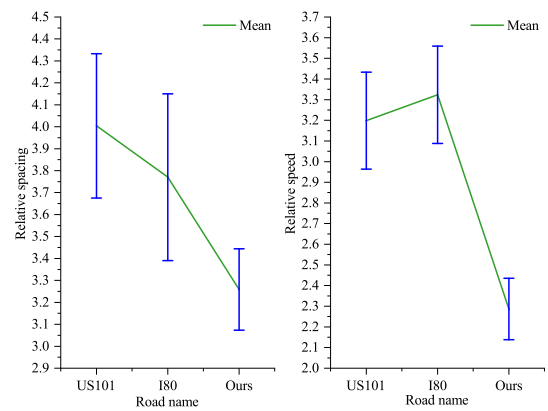

Fig. 3. Threshold selection for different roads.

constructed in our study. We analyze the data on relative distance and relative speed to the preceding vehicle when initiating lane changes for three road conditions, after transforming the data into a normal distribution. The statistical analysis is shown in Fig. 3. It reveals different means and 95% confidence interval widths under three distinct road conditions, thereby confirming that different roads can lead to varying thresholds for priority triggering. Specifically, the mean relative distance to the preceding vehicle after transformation differed by 0.75 between US101 and our simulated road environment when starting to change lanes, and the relative speed differed by 0.91. Compared to our simulated condition, the I80 road shows a difference of 0.51 in the mean relative distance and a difference of 1.04 in relative speed. These discrepancies in thresholds are primarily attributed to differences in traffic flow resulting from the unique road structure of each road.

2) Priority Detection and Triggering Mechanism Module: To demonstrate the practical utility of our designed priority detection and dynamic event-triggered mechanism in mitigating conflicts that may arise during multi-vehicle collaborative lane-changing overtaking decisions, this study records the speeds and TTC values of three CVs in two scenarios: with and without this module. During the model training phase, we select the episode with the highest rewards from the last 100 episodes and record the state data of CVs in this episode. Considering the decision-making process of CVs spans multiple time intervals,

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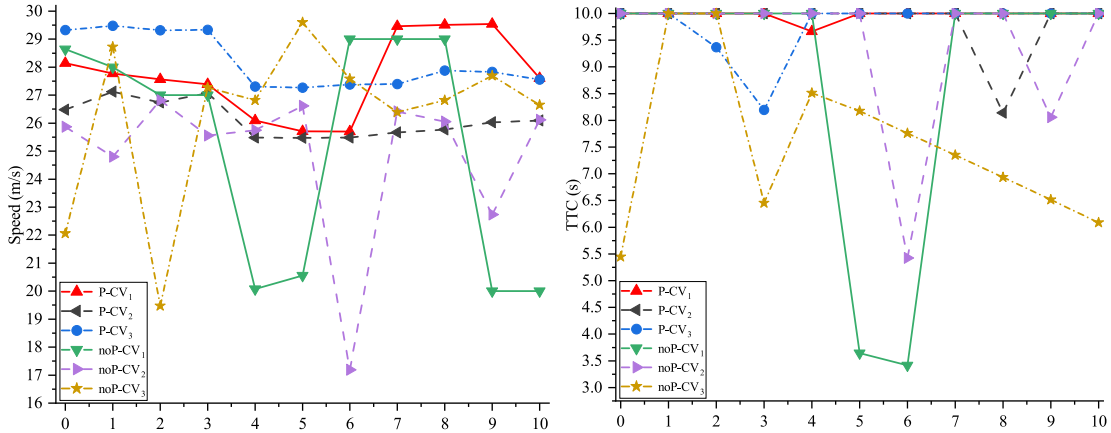
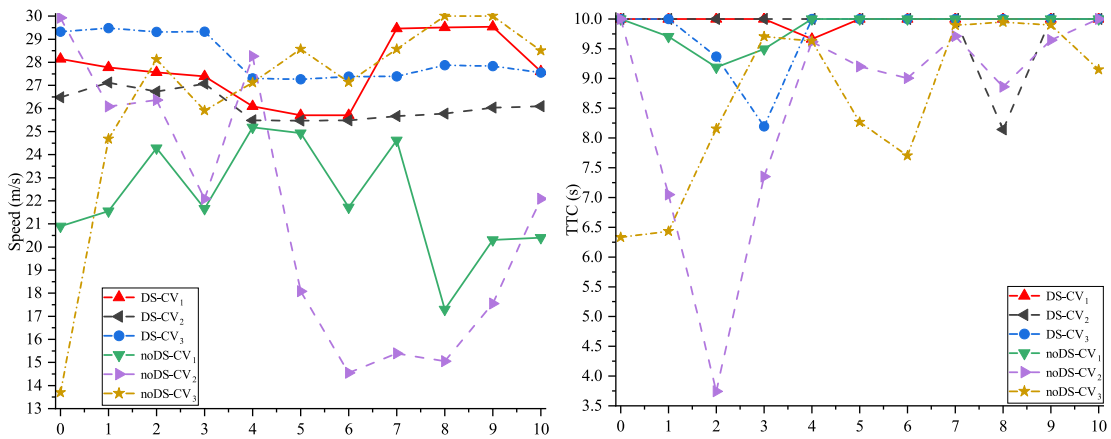


(a) Speed change process (b) TTC change process

Fig. 4. Performance comparison of priority module.



(a) Speed change process (b) TTC change process

Fig. 5. Performance comparison of driving style module.

we choose 11 time points to clearly illustrate the changes in speed and TTC of CV, with each time point representing the average value for that interval. $P\text{-}CV_1$, $P\text{-}CV_2$, and $P\text{-}CV_3$ represent the CVs incorporating the priority detection and dynamic event-triggered mechanism, whereas $noP\text{-}CV_1$, $noP\text{-}CV_2$, and $noP\text{-}CV_3$ represent the CVs without this integration. Fig. 4 shows that $noP\text{-}CV_1$, $noP\text{-}CV_2$, and $noP\text{-}CV_3$ experience more significant fluctuations in speed and TTC, especially $noP\text{-}CV_1$ between 4-6 time intervals, where competitive conflicts led to notable decreases in speed and TTC, reducing TTC to 3.42 and posing a significant safety risk to surrounding vehicles. This phenomenon indicates that integrating the priority detection and dynamic event-triggered mechanism enhances collaboration among CVs. By evaluating the operational states of CVs to calculate the opi, this mechanism can determine which CV has higher priority when two CVs encounter resource competition based on this opi, allowing that CV to make decisions first, thereby flexibly avoiding significant fluctuations and decreases in speed of CV and TTC due to potential decision-making conflicts.

3) Driving Style Module: In the MCCV scenario, the variation in the driving styles of surrounding vehicles significantly affects vehicle safety. To delve deeper into the impact of vehicle driving styles on lane-changing overtaking decisions, this study designs a set of experiments aimed at evaluating the efficiency of lane-changing overtaking maneuvers by CVs, with and without consideration of driving style. This module inherits and applies the data processing and plotting methods described in Section V-B2. $DS\text{-}CV_1$, $DS\text{-}CV_2$, and $DS\text{-}CV_3$ represent CVs integrated with the driving style module, while $noDS\text{-}CV_1$, $noDS\text{-}CV_2$, and $noDS\text{-}CV_3$ indicate CVs without this integration. Fig. 5 shows that $DS\text{-}CV_1$, $DS\text{-}CV_2$, and $DS\text{-}CV_3$ exhibit more stable driving speeds and TTC compared to $noDS\text{-}CV_1$, $noDS\text{-}CV_2$, and $noDS\text{-}CV_3$, which demonstrate less stable speeds and larger TTC fluctuations, particularly $noDS\text{-}CV_2$ whose TTC drop to 3.74 s, significantly increasing

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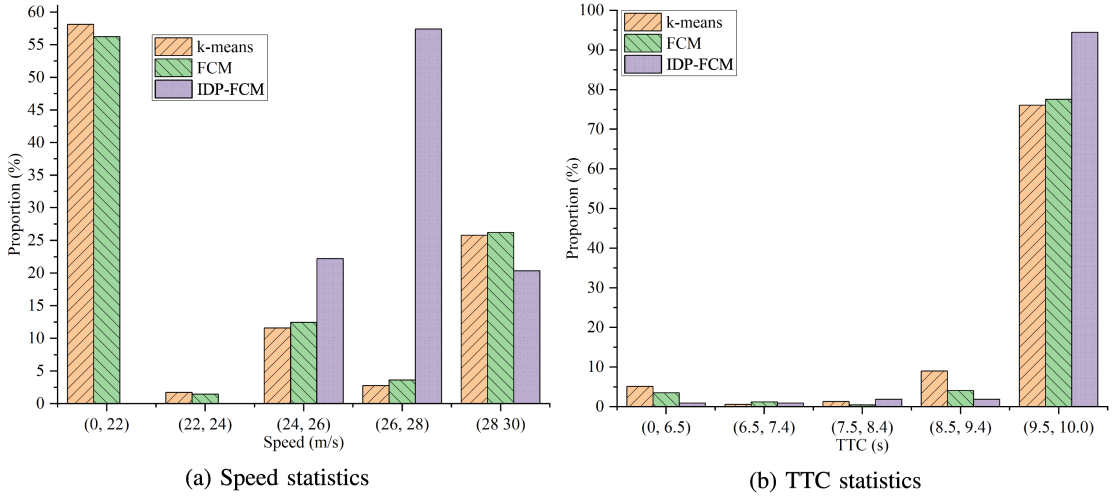

Fig. 6. Performance comparison of the IDP-FCM algorithm.

collision risk. This phenomenon highlights that traditional TTC calculation methods [26], which only consider relative distance and speed, may underestimate actual safety risks by overlooking the differences in driving styles between autonomous and human-driven vehicles in the MCCV scenario, due to varying safety standards and driving habits. Our proposed IDP-FCM algorithm effectively identifies different driving styles and optimizes lane-changing overtaking strategies accordingly to reduce safety risks. Experimental results confirm the effectiveness of our method in recognizing driving style differences and enhancing the speed and safety of CVs in executing lane-changing overtaking decisions.

4) Performance Comparison of the IDP-FCM Algorithm: To validate the superiority of our proposed IDP-FCM algorithm in handling complex driving style recognition in MCCV environments, this study designs a set of comparative experiments to evaluate the performance differences in vehicle driving style identification between the IDP-FCM algorithm and the existing k-means and FCM algorithms [11], [27]. In the experiments, we divide the data range into five intervals based on speed and TTC and conduct a comprehensive analysis of the speed and TTC data of three CVs after aggregation. The results demonstrating the impact of driving styles on overall driving decisions obtained through these three algorithms are shown in Fig. 6. While the speed distribution (28, 30) is slightly lower than that of k-means and FCM algorithms, within the speed distribution (26, 28), the IDP-FCM algorithm showed an increase in the proportion of 54.7% and 53.8% compared to k-means and FCM algorithms, respectively; and in the safest TTC interval (9.5, 10), the performance of the IDP-FCM algorithm exceeded that of k-means and FCM algorithms by 18.4% and 16.9%, respectively. Moreover, in the most unsafe interval (0, 6.5), the IDP-FCM algorithm demonstrates a lower proportion compared to the two benchmark algorithms, reducing by 4.1% and 2.6% respectively. These results sufficiently prove the significant advantages of the IDP-FCM algorithm in enhancing the speed decisions and driving safety of CV. The superiority of the IDP-FCM algorithm over k-means and FCM algorithms is mainly attributed to its incorporation of interquartile range outlier treatment, the k-means algorithm for initializing cluster centers, and improvements in online learning techniques. Employing the IQR method to remove outliers, could reduce the interference of noise data; using the k-means algorithm for the initial selection of cluster centers enhances the algorithm's stability and convergence speed; the application of online learning techniques enables IDP-FCM to update cluster information in real-time, adapting to the dynamic changes in driving behavior. Through the comprehensive application of these methods, the adaptability and accuracy of the IDP-FCM algorithm in dynamic driving scenarios are greatly enhanced.

5) Contrastive Learning Module: The integration of contrastive learning in our proposed MCPDDQN algorithm significantly accelerated the model's convergence process and improved its stability, thus further optimizing the model's decision-making efficiency. This advancement is demonstrated through comparative experiments: one set utilized contrastive learning (marked as CL), while the other did not (marked as $noCL$). We analyze the performance improvements brought by the integration of the contrastive learning module from three perspectives: average reward, speed statistics, and TTC statistics, with the results shown in Fig. 7. Fig. 7(a) shows the change of average reward with the number of training episodes, indicating that the CL method achieved convergence by the 800th iteration, whereas the $noCL$ method only reached this state by the 1680th iteration. Moreover, in terms of the stability of the average reward function obtained from each episode of training, the CL method exhibited significantly less fluctuation than the $noCL$ method, indicating improved stability throughout the experimental process. Additionally, the CL method's average reward value at final convergence is approximately 0.8 higher than that of the $noCL$ method. Fig. 7(b) demonstrates the impact of the contrastive learning module on speed. It can be seen that the CL method has a higher proportion of high speeds compared to the $noCL$ method, especially in the lower speed range (0, 24), where the $noCL$ method accounts for 29.53%, while the CL method accounts for 0%. This indicates a significant improvement in

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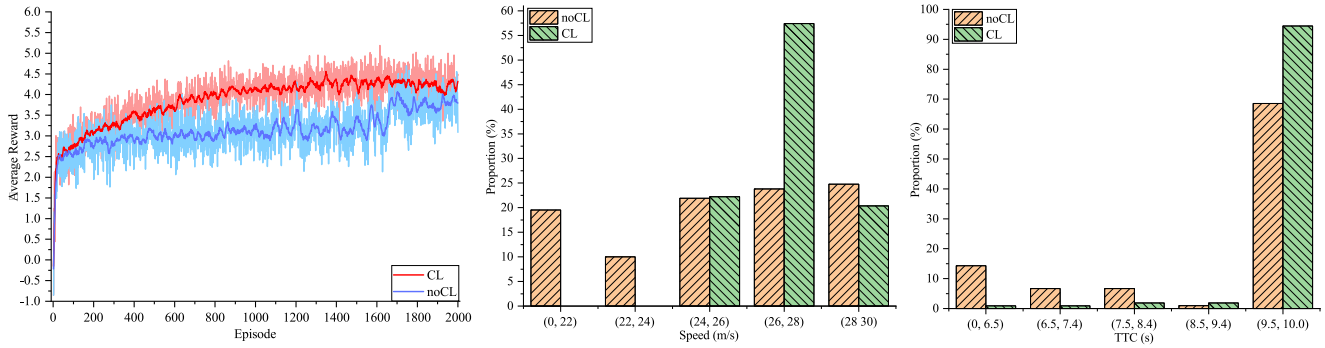
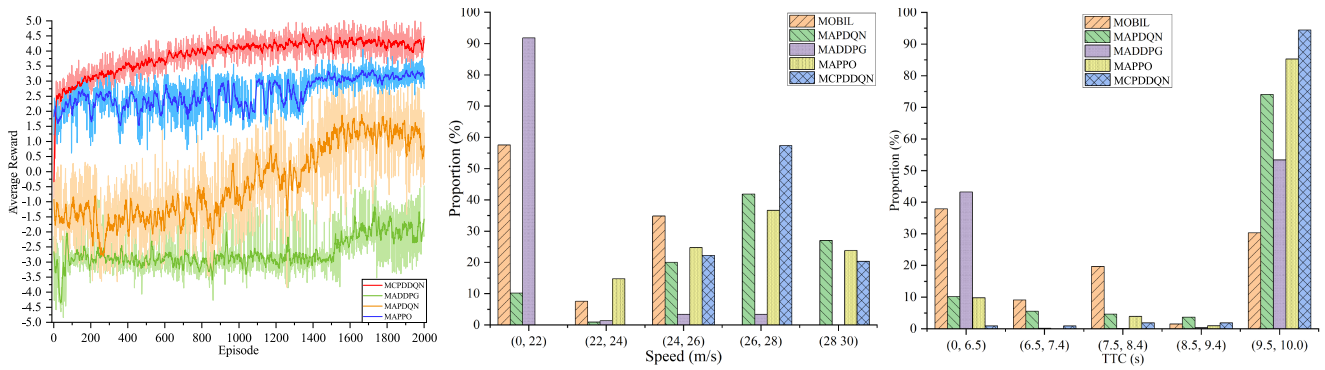


(a) Average reward (b) Speed statistics (c) TTC statistics

Fig. 7. Performance comparison of contrastive learning module.



(a) Average reward (b) Speed statistics (c) TTC statistics

Fig. 8. Performance comparison of different algorithms.

vehicle efficiency with the integration of contrastive learning. For the safety metric (i.e., TTC), in the safest range (9.5, 10), the CL method shows a 25.87% higher TTC proportion compared to the $noCL$ method, and in the least safe range (0, 6.5), the CL method shows a 13.36% lower TTC proportion than the $noCL$ method. These performance improvements are attributed to contrastive learning's deepening of the model's understanding of the differences between effective and non-ideal behaviors by constructing positive and negative samples from randomly sampled data in the experience pool and calculating contrastive loss. In this way, contrastive learning reinforces the model's adaptability to complex environments and enhances its ability to identify unreasonable decisions, thereby maintaining efficient and stable performance under various environmental conditions.

6) Comparative Algorithms: To substantiate the superiority of our proposed MCPDDQN algorithm, we conduct a set of comparative experiments including the following algorithms as benchmarks:

- MAPDQN: Considering our algorithm as an enhancement of the PDQN algorithm [28], this study extends the PDQN algorithm to a multi-agent environment, creating the new MAPDQN algorithm.
- MAPPO [23]: It utilizes a clipping ratio function to optimize objectives, dynamically selects clipping parameters, and introduces intrinsic rewards to enhance performance.
- MADDPG [29]: It combines centralized training with decentralized execution, where each agent makes decisions independently based on local observations, with global information used to enhance performance.
- MOBIL [18]: A micro-traffic model-based lane-changing evaluation model that considers longitudinal acceleration to assess the utility and risk of lane changes.

The results of the comparative experiments, as shown in Fig. 8, indicate that our MCPDDQN algorithm significantly outperforms the MAPDQN, MAPPO, and MADDPG algorithms in terms of average rewards, with improvements of 1.6, 3.0, and 6.0, respectively. Additionally, the MCPDDQN algorithm demonstrated the fastest convergence speed, reaching convergence after approximately 800 iterations, whereas other algorithms required between 1300 and 1600 iterations. In guiding the speeds of CVs, we adopt the data processing method described in Section V-B2. As seen in Fig. 8(b), the MCPDDQN algorithm maintains the speeds of CVs within the interval (24,30), whereas other algorithms result in lower speed distributions, particularly the MADDPG algorithm and MOBIL method, which occupy 91.8% and 57.6% of the lowest speed interval (0,22), respectively. On the TTC metric, the MCPDDQN algorithm had the highest proportion in the safest interval $(9.5, 10)$, reaching 94.4%, and the lowest proportion in the most unsafe interval $(0, 6.5)$, at just 0.93%, highlighting its significant driving safety.

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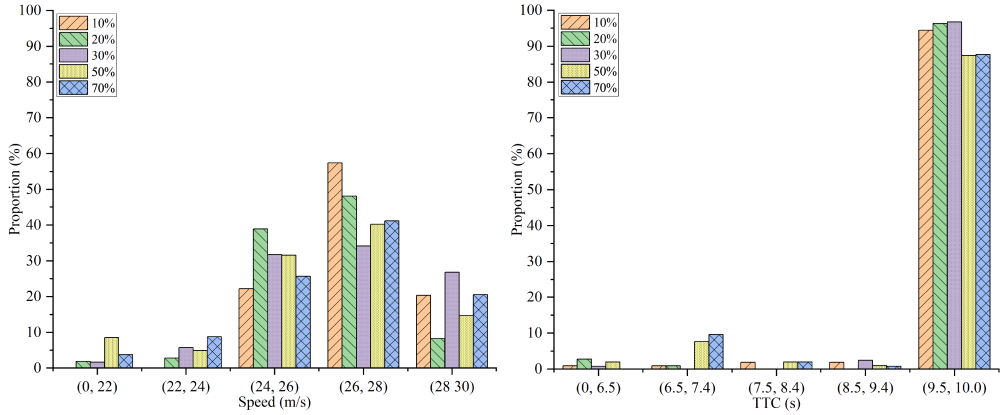

(a) Speed statistics (b) TTC statistics

Fig. 9. Performance comparison at different penetration rates of the CV.

The significant advantage of the MCPDDQN algorithm over the MAPDQN and MAPPO algorithms is mainly due to its actor-critic architecture, which leverages the benefits of both PPO and DDQN. The actor network optimizes the selection of continuous actions using PPO's efficient policy gradient optimization method to reduce sample correlation and variance, thereby enhancing the accuracy and learning rate of action selection. The critic network evaluates discrete actions and their corresponding continuous action values using DDQN, applying the Dueling DQN architecture to separate state value and action advantage functions, improving learning efficiency and decision quality. Compared to the MADDPG algorithm, by parameterizing the action space, it effectively integrates discrete and continuous actions, enabling the algorithm to consider both lateral lane changes and longitudinal speed adjustments in lane-changing and overtaking tasks, ensuring the rationality and efficiency of action selection. Conversely, the MADDPG algorithm treats discrete actions discretely, leading to a decrease in decision accuracy. The MOBIL model relies on fixed rules to evaluate lane-changing decisions, lacking adaptability to complex dynamic environmental changes, which results in insufficient performance and flexibility in varying traffic scenarios. Additionally, the MCPDDQN algorithm integrates contrastive learning, optimizing positive and negative sample generation strategies to enhance model stability and convergence speed, demonstrating its rapid learning and adaptability in multi-agent collaborative tasks.

7) Impact of Different Connected Vehicle Penetration Rates: To validate the performance of the proposed algorithm in environments with different CV penetration rates, we conduct additional experiments varying the rates at 10%, 20%, 30%, 50%, and 70%. The results are presented in Fig. 9, illustrating the speed and TTC statistics under these conditions. It should be noted that the previous experimental analyses are based on a 10% penetration rate of CVs. As shown in Fig. 9(a), the speed distribution of CVs varies across different penetration rates. At a 10% penetration rate, most vehicles operate in the (26, 28), with about 60% in this interval. As the penetration rate increases to 20%, the distribution evens out between the (24, 26) and (26, 28), reflecting improved traffic flow and coordination. At a 30% penetration rate, the majority of CVs are still concentrated in the (24, 26) and (26, 28), but there is an increase in the (28, 30), indicating enhanced traffic efficiency. Interestingly, at 50% and 70% penetration rates, while there is a balanced distribution across the (24, 26), (26, 28), and (28, 30), the proportion of CVs in the highest speed range (28, 30) slightly decreases compared to lower penetration rates. This suggests that while overall traffic flow remains efficient, the increased interactions and coordination required among more connected vehicles lead to more frequent speed adjustments, slightly reducing the highest speeds achieved. Fig. 9(b) shows that at all penetration rates, the majority of CV maintain a TTC in the (9.5, 10.0), suggesting that most CVs are maintaining safe distances from each other. At a 10% penetration rate, most vehicles maintain a high TTC, demonstrating that our proposed method ensures safety under low penetration conditions. As the penetration rate increases to 20% and 30%, the proportion of CVs in the lower TTC intervals decreases, indicating improved safety as more vehicles can communicate and anticipate each other's actions better. At 50% and 70% penetration rates, the proportion of CVs in the safest TTC interval (9.5, 10.0) remains high but shows a slight decrease compared to 30%. This slight reduction is likely due to the increased complexity of managing a larger number of CVs, leading to more frequent interactions and adjustments in speed and distance to maintain optimal flow and safety. These findings validate the robustness of our collaborative overtaking strategy across various CV penetration scenarios.

VI. Conclusion

This paper proposes a novel collaborative overtaking strategy for the MCCV scenario, significantly enhancing the safety and efficiency of traffic. By implementing collaborative lane-changing overtaking strategies, introducing priority detection and event-triggered mechanisms, and developing a dynamic algorithm to capture vehicle driving styles, we effectively address potential competition and conflicts among connected vehicles, reduce computational resource consumption, and improve

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algorithm stability. Notably, the novel priority detection and triggering method optimizes collaboration among connected vehicles, and the IDP-FCM algorithm successfully addresses the safety challenges brought by the diversity of driving styles. Furthermore, our proposed multi-agent based MCPDDQN algorithm, integrating contrastive learning strategies, solves the stability issues in distributed multi-agent reinforcement learning and provides reliable decision support for real-time driving tasks. For our future work, we will explore the generalization of these strategies in different traffic environments globally, considering the differences in traffic regulations, road conditions, and driving habits across countries and regions, to achieve global adaptability of the strategy.

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